

Andrey Simachev

Senior Front-End Engineer and Architect

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Professional Summary

Senior front-end engineer and architect with over 15 years building production web applications, designing component architectures, and leading cross-functional engineering teams. Founding engineer of an open-source, local-first knowledge-management platform used by thousands of people. For the past year, working AI-natively: Claude Code as a primary development tool, spec-driven workflows as the default, and agentic multi-step tasks where I orchestrate subagents instead of writing every line by hand. That shift is how I rebuilt the Anytype desktop client end-to-end as a solo effort. Strong product sense: designed and shipped analytics systems that directly informed feature prioritization, and consistently aligned engineering decisions with user and business outcomes.

Technical Skills

Languages and core: TypeScript (strict), JavaScript (ES2023+), HTML, CSS, accessibility, cross-browser compatibility, responsive design

UI frameworks and libraries: React 19, MobX, MobX-React-Lite, Lexical, shadcn/ui, Tailwind CSS v4, Electron, Tauri v2

Visualization and graphics: D3.js, PixiJS, force-directed graphs, interactive canvas and SVG rendering, WebGL, custom layout engines

Build, runtime and tooling: Vite, Bun (runtime, package manager, test runner), Node.js, webpack and rspack migration experience, ESLint, Prettier

APIs and data layer: gRPC, Protocol Buffers (ts-proto), REST, WebSockets, PostgreSQL, Greenplum, MongoDB, SQLite (bun:sqlite)

Desktop and cross-platform: Electron, Tauri v2, Rust (Tauri shell, commands and plugins), native packaging and release engineering for macOS, Windows and Linux. Electron Builder, Electron Forge, Electron Packager, Squirrel auto-update, code signing and Apple notarization. Tauri bundler, DMG, NSIS, MSI, AppImage, deb and rpm installers.

AI-native development: Claude Code, Cursor, spec-driven development, agentic coding workflows, subagent orchestration, prompt engineering, AI-driven large-scale migrations and refactors, custom skills and hooks for repeatable workflows

Quality and DevOps: Git, CI/CD pipelines, code review, unit and integration testing, release engineering

Product and process: Agile, Scrum, product analytics, roadmap planning, hiring, mentoring, technical leadership

Professional Experience

Senior Front-End Engineer and Architect

Anytype.io, Berlin, Germany. Apr 2021 to Present.

Founding engineer of an open-source, local-first knowledge-management platform. Built the core front-end codebase from the ground up and grew an active open-source contributor community around it.

- Shipped core product surfaces used by thousands of users: a block-based rich-text editor with 20+ block types and drag-and-drop composition, a relational database UI with seven view types (grid, gallery, kanban, calendar, graph, list, timeline), an interactive knowledge graph built on D3.js force-directed layout, a real-time collaborative chat system with messages, reactions, replies, mentions, file attachments and drag-and-drop uploads, a notification system, and a Chrome extension.
- Architected the front-end data layer: reactive MobX stores consuming a gRPC event stream from a Go middleware, with local caching, optimistic updates and conflict resolution. Owned the client-side contract with backend as the middleware surface evolved.
- Designed and shipped an internal product analytics system that tracked feature adoption and user engagement, directly informing roadmap prioritization.
- Owned release engineering and CI/CD pipelines across macOS, Windows and Linux. Handled cross-platform packaging, code signing and Apple notarization for Electron and Tauri builds.
- Contributed to product roadmap and sprint prioritization in an Agile workflow, aligning technical architecture with user needs and business goals across a cross-functional team.
- Mentored front-end engineers and contributed to hiring, code reviews and sprint planning.

AI-native rewrite, anytype-bun, 2025 to present. Leading a ground-up rewrite of the desktop client, executed as a mostly-solo effort through spec-driven development with Claude Code. The workflow: I write detailed specs and plans as the primary artifact, treat the model as a collaborator that reads and implements against those specs, orchestrate subagents for parallel research and isolated changes, and review every diff. It has changed my sense of what one senior engineer can realistically own.

Full stack and toolchain modernization:

- **Runtime:** Electron to Tauri v2, with smaller binaries, a native Rust shell and tighter OS integration.
- **Editor:** custom contentEditable implementation to Lexical, rebuilt with the full block model on top.
- **Styling:** Sass to Tailwind CSS v4.
- **Build system:** webpack to Vite and Bun, using Bun as package manager, runtime and test runner.
- **State and UI:** React 19, TypeScript strict, MobX, MobX-React-Lite.
- **Transport:** full protobuf layer rewrite from google-protobuf to ts-proto, over gRPC event streaming.
- **Packaging:** cross-platform Tauri bundler pipeline with DMG, MSI and AppImage targets, replacing the legacy Electron Builder setup.
- **Features rebuilt end-to-end:** the chat system with real-time messages, reactions, replies, mentions, attachments and file drag-and-drop, and the block editor on Lexical.

Earlier AI-driven work on the legacy client, anytype-ts. Led a series of large automated migrations through Claude Code: jQuery removal across the codebase, DOM utility unification, a full MobX-React-Lite conversion, and the rspack to Vite and Bun build

migration. Each was framed as a spec the model executed against, not a sequence of hand-written edits. Ongoing feature development, community-feedback work and bug fixes alongside.

Tech: React 19, TypeScript strict, MobX, Lexical, Tailwind CSS v4, Tauri v2, Rust, Electron, Electron Builder, Vite, Bun, gRPC, Protocol Buffers, ts-proto, D3.js, PixiJS, Node.js, Claude Code, Cursor, Git.

Chief Technology Officer

devtodev, Vilnius, Lithuania. Oct 2018 to Apr 2021.

Led engineering for a SaaS product-analytics platform serving game developers, managing a cross-functional team of 15 engineers in an Agile environment.

- Authored technical specifications and roadmaps, aligning product development with business objectives across departments.
- Introduced a Greenplum database cluster, reducing complex analytical query times by more than 60 percent.
- Oversaw a ground-up platform redesign that improved user retention and reduced support-ticket volume by 40 percent.
- Built and grew the front-end team: hiring, one-on-ones, performance conversations, mentoring and setting technical direction.

Tech: React, TypeScript, MobX, Sass, Node.js, PostgreSQL, Greenplum, MongoDB, Git.

Senior Developer

Game Insight, Vilnius, Lithuania. Dec 2010 to Oct 2018.

- Developed the backend and CMS for Narr8, an interactive storytelling app with 500,000+ downloads.
- Built the front-end UI layer for Tank Domination, an online multiplayer game.
- Created internal productivity tools, including a Slack integration framework and an analytics dashboard used across the studio.
- Architected talkover.com, a real-time social chat platform handling concurrent users, giving me early experience with the kind of real-time messaging work I later brought into Anytype.

Tech: Node.js, JavaScript, HTML, CSS, MongoDB.

Earlier Experience

Astrum Online Entertainment, Moscow, Russia. 2004 to 2010. Full-stack engineer on browser-based MMO games, game UIs, server-side logic and back-office tooling.

Education

Bachelor of Science in Information Security National Research University MPEI, Moscow, Russia. 2011 to 2015.